

SHORT BIO. During my Ph.D., I focused on *causal decision-making* (NeurIPS-25, UAI-26a, ICLR-26) and *causal effect identification* (NeurIPS-24, UAI-26c). I investigated how to formally model diverse real-world problems within a causal framework and derive optimal decisions from such structured representations.

In particular, I developed methods for robust decision-making under uncertainty (NeurIPS-25), domain shifts (UAI-26a), and even counterfactual actions (ICLR-26).

EDUCATION

Graduate School of Data Science, Seoul National University

Seoul, Republic of Korea

Ph.D. in Data Science

Sep. 2022 – Feb. 2027

Advisor: Prof. Sanghack Lee.

Thesis: *Generalizing Causal Decision-Making: Markov Equivalence, Transportability and Counterfactuals*.

Doctoral Qualifying Exam waived due to outstanding academic performance.

Department of Mathematics and Statistics, Sejong University

Seoul, Republic of Korea

B.S. in Applied statistics and Mathematics (Double major), *Cum Laude*

Mar. 2016 – Aug. 2022

Advisors: Prof. Ji Eun Lee and Jin Hee Yoon.

(2.5 years military service)

RESEARCH EXPERIENCE

Samsung Research | AI Research Engineer

Starting in Mar. 2027

Furiosa AI | AI Research Engineer Internship

Jun. 2026 – Sep. 2026

Causality Lab, Seoul National University | Ph.D. Researcher

Jan. 2023 – Feb. 2027

Functional Analysis Lab, Sejong University | Undergraduate Researcher

Jul. 2021 – Dec. 2022

ISDS Lab, Sejong University | Undergraduate Researcher

Jan. 2021 – Dec. 2022

PUBLICATIONS

* for joint authorship and [†] for corresponding author.

- ◇ [On Transportability for Structural Causal Bandits](#).

Min Woo Park and Sanghack Lee[†]. *Uncertainty in Artificial Intelligence* (UAI), 2026.

- ◇ [Breaking Bad: Component-wise Parent Deletion for Score-Based Causal Discovery](#).

Min Woo Park^{*}, Taehui Yun^{*}, Youngin Jang, Younseok Yeom, Jonghwan Kim, Jiyeon Kang, Songseong Kim, Hyemin Jung, Sangmin Lee, Jongseong Jang[†], and Sanghack Lee[†]. *Uncertainty in Artificial Intelligence* (UAI), 2026.

- ◇ [Canonical Domain Reduction for Partial Counterfactual Identification](#).

Yesong Choe, Yeahoon Kwon^{*}, **Min Woo Park**^{*}, and Sanghack Lee[†]. *Uncertainty in Artificial Intelligence* (UAI), 2026.

- ◇ [Counterfactual Structural Causal Bandits](#).

Min Woo Park and Sanghack Lee[†]. *International Conference on Learning Representations* (ICLR), 2026.

- ◇ [Structural Causal Bandits under Markov Equivalence](#).

Min Woo Park, Andy Ardit, Elias Bareinboim[†], Sanghack Lee[†]. *Neural Information Processing Systems* (NeurIPS), 2025

CausalAI Laboratory, Columbia University, Technical Report (R-122).

- ◇ [Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference](#).

Yonghan Jung[†], **Min Woo Park**, and Sanghack Lee[†]. *Neural Information Processing Systems* (NeurIPS), 2024.

- ◇ [Computation of the iterated Aluthge, Duggal, and Mean transforms](#).

Min Woo Park^{*}, and Ji Eun Lee^{*†}. *Filomat*, 2023.

- ◇ Convergence of the iterated mean transforms of a 2×2 matrix

Ji Eun Lee^{*†} and **Min Woo Park**^{*}. under review.

AWARDS AND HONORS

- NRF Ph.D. fellowship in science and engineering, 2026** | National Research Foundation of Korea
▷ *Active Learning-Based Action Space Optimization for Causal Multi-Armed Bandits under Incomplete Information.* Sep.2026 – Aug.2027
- UAI 2026 scholarship** Aug. 2026
- Graduate Student Instructor (GSI) scholarship** Spring 2024, Spring 2025, Spring 2026
- BK21 FOUR scholarship** Fall 2024, Fall 2025
- First place merit scholarship** | Sejong University Fall 2020, Fall 2021
- Fourth place award** | The 9th College of Natural Sciences Academic Conference, Sejong University
▷ *Mitigating Spurious Regression in Time-Series Data via Granger Causality.* Nov. 2021
- Third place award** | The 8th College of Natural Sciences Academic Conference, Sejong University
▷ *The Significance of the Euler's Number and Its Applications.* Nov. 2020

PROJECTS

- Towards More Scalable Score-based Causal Discovery** | LG AI Research Sep. 2025 – Aug. 2026
▷ Leading a project on scalable score-based causal discovery algorithms.
- Developed a novel operator enabling score-based causal discovery algorithms to escape local optima.
- Investigating parallelization strategies for score-based causal discovery algorithms.
▷ Resulting paper accepted at *Uncertainty in Artificial Intelligence (UAI-26b)*.
- Software available at <https://github.com/LGAI-Research/breaking-bad>.
▷ Korean patent application filed.
- Hyundai NGV Data Boot-Campus** | Hyundai Motor Company
▷ **2025.** Developed high-voltage battery performance prediction technology. Jul. 2025 – Sep. 2025
▷ **2024.** Developed an intelligent system for predicting vulnerable regions of automotive body panel stiffness using differential geometrybased automatic curvature analysis. Jul. 2024 – Sep. 2024
- Optimal Mixed Policies Completeness** | Oxford University Aug. 2023 – Feb. 2024
▷ Studied characterization of conditions under which an intervention effect of a variable is positive.
▷ Acknowledged for helpful comments in *Toward a Complete Criterion for Value of Information in Insoluble Decision Problems*. *Transactions on Machine Learning Research (TMLR)*, 2024.

TEACHING ASSISTANTS

- Causal Inference for Data Science** Spring 2024, Fall 2024, Spring 2025, Spring 2026
▷ Topics included Bayesian networks and graphical causal inference.
- Machine Learning and Deep Learning for Data Science II** Fall 2023
▷ Covered kernel methods, Gaussian processes, and related topics in machine learning.
- Basic Mathematics for Artificial Intelligence (Undergraduate)** Spring 2022
▷ Covered linear algebra, basic statistics, and probability theory.

ACADEMIC SERVICES, PROFESSIONAL TALKS AND POSTERS

Academic Services

- Conference Reviewer: **2027 ICLR**, **2026 NeurIPS**, UAI, ICLR, ICML, **2025 ICML**
Reviewed AI/ML lecture video content from LG AI Research for LG Aimers
Summer 2025, Winter 2025, Summer 2026

Invited Talks

- Major Exploration for Sejong University Students | Seoul, Republic of Korea Nov. 2026 (scheduled)
SNU GSDS student research seminar | Seoul, Republic of Korea Dec. 2024, Apr. 2025

Poster Sessions

UAI 2026 poster session Amsterdam, Netherlands	Aug. 2026
ICLR 2026 poster session Rio de Janeiro, Brazil	Apr. 2026
ExploreCSR poster session Seoul, Republic of Korea	Dec. 2025
NeurIPS 2025 poster session San Diego, USA	Dec. 2025
JKAIA 2025, NeurIPS preview session Seoul, Republic of Korea	Nov. 2025
SNU GSDS workshop poster session Jeju, Republic of Korea	Jul. 2025
NeurIPS 2024 poster session Vancouver, Canada	Dec. 2024

REFERENCE

Sanghack Lee | Graduate School of Data Science, Seoul National University sanghack@snu.ac.kr
Associate professor

Ji Eun Lee | Department of Mathematics and Statistics, Sejong University jieunlee7@sejong.ac.kr
Associate professor

Jin Hee Yoon | Dept. of AI and Information Technology, Sejong University jin9135@sejong.ac.kr
Associate professor

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